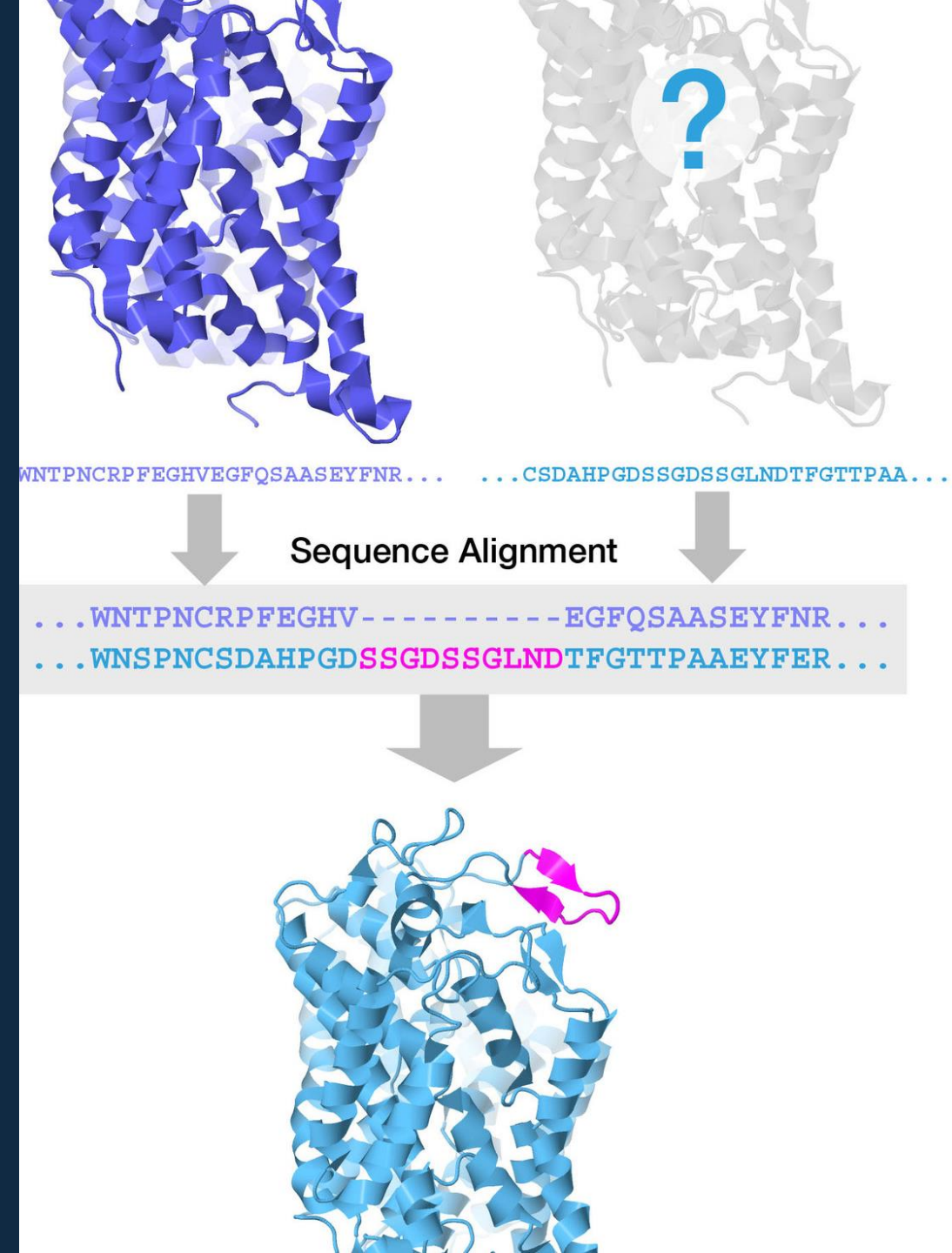


AI-Assisted Template-Based Protein Structure Modeling

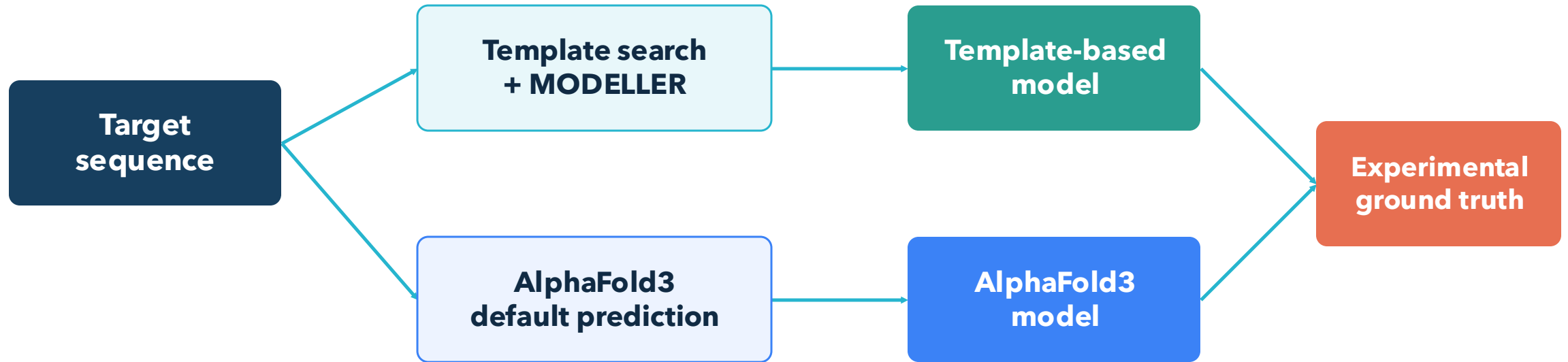
Project 2 Plan · CMP_SC 8170

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October 2, 2026



Build two AI-assisted agents that predict and evaluate protein structures for three CASP15 targets.



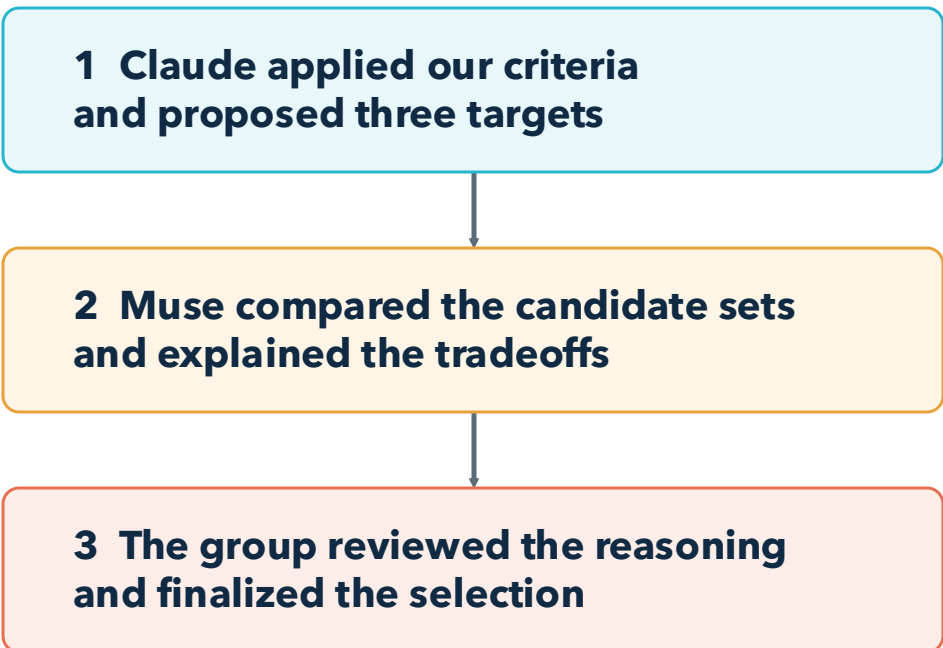
Research question

How does template-based modeling perform across easy, intermediate and hard targets compared with AlphaFold3?

Criteria defined by our group

- 1 Official CASP15 protein target
- 2 One chain modeled consistently
- 3 One easy, intermediate and hard case
- 4 Fewer than 500 amino acids
- 5 Public experimental structure
- 6 One clear evaluation unit
- 7 Manageable within one week

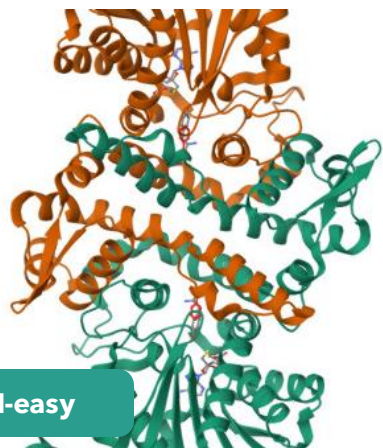
AI-assisted decision process



AI supported the decision; the group supplied the criteria and approved the final targets.

Selected CASP15 targets

04



T1124

384 aa • PDB 7UX8

EU: residues 7-384

Strong-template baseline



T1123

266 aa • PDB 7UZZ

EU: residues 33-258

Intermediate template signal



T1127

211 aa • PDB 8XBP

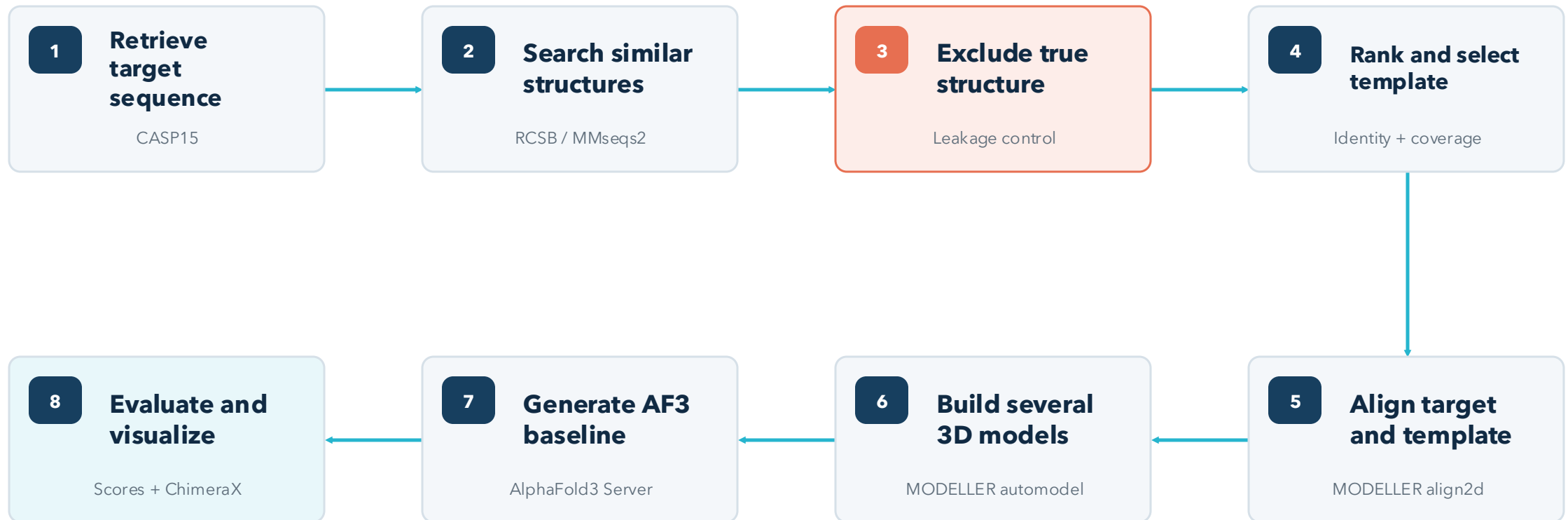
EU: residues 6-210

Weak-template stress test

All three use one evaluation unit and form a manageable size gradient: 211 → 266 → 384 residues.

End-to-end workflow

05

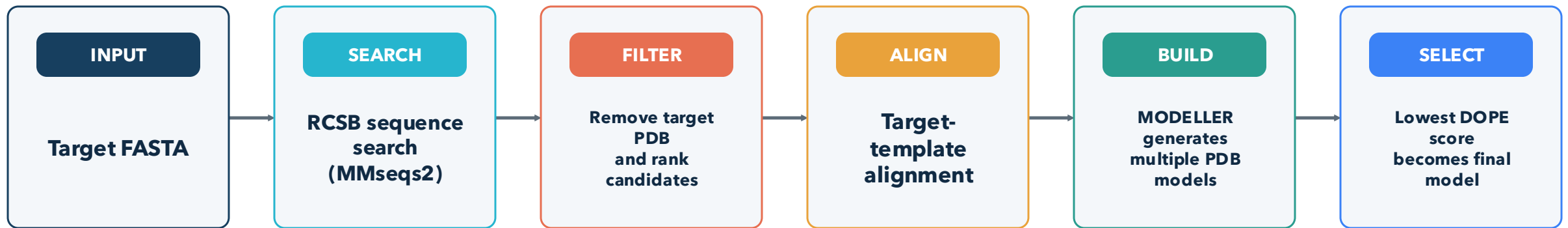


Prediction uses no ground-truth coordinates. The true structure appears only during evaluation.

Agent 1: template-based model building

06

Claude coordinates the workflow; scientific tools perform the database search and coordinate calculations.



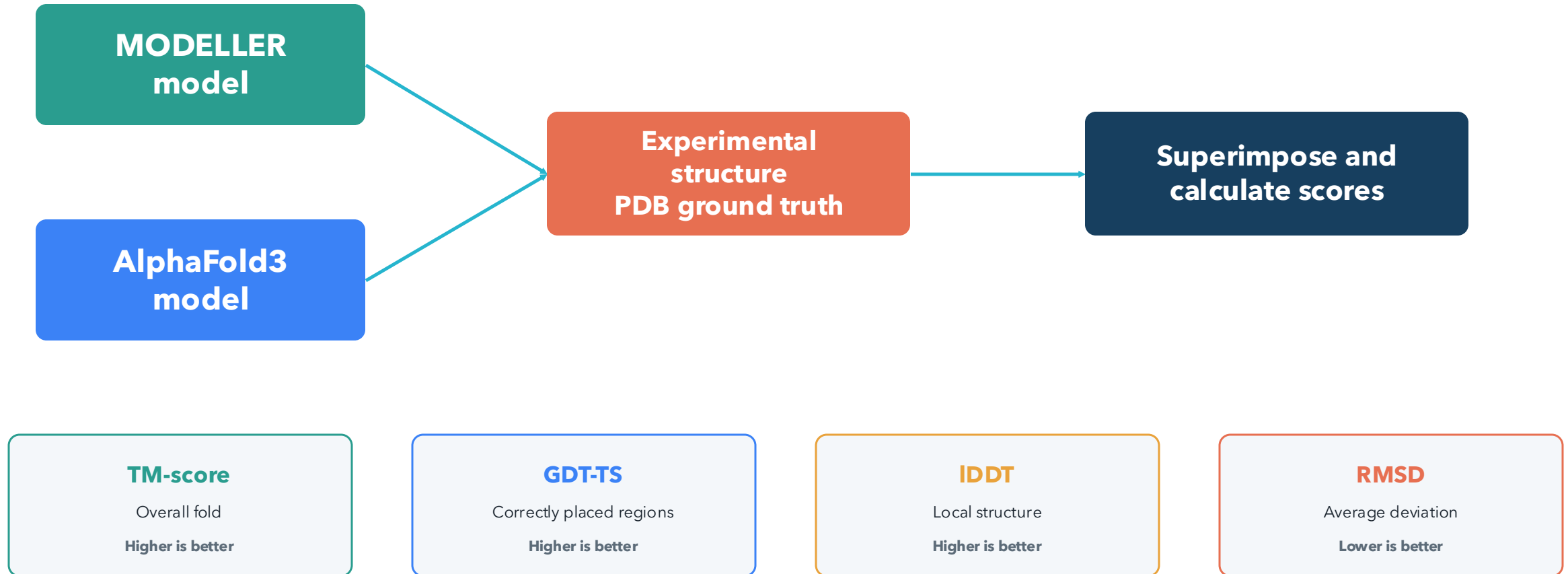
Template ranking signals

Sequence identity · Target coverage · E-value · Experimental resolution · Missing residues

The agent records every template considered and explains why it selected the final one.

Agent 2: evaluation against ground truth

07

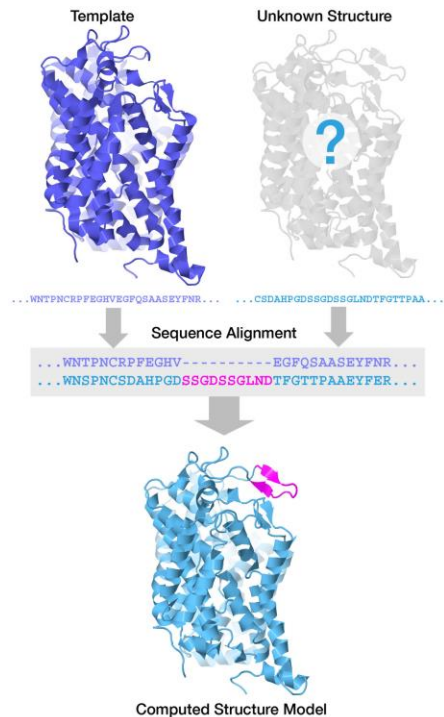


Stage	Selected resource	Purpose
Targets	CASP15	Sequences, classifications and evaluation units
Template database	RCSB PDB	Experimentally determined structures
Template search	RCSB Search / MMseqs2	Find similar proteins with known structures
Alignment + modeling	MODELLER	Align target to template and build 3D models
Baseline	AlphaFold3 Server	Generate the comparison prediction
Evaluation	TM-score / TM-align + IDDT	Superimpose structures and calculate metrics
Visualization	ChimeraX	Display structures and colored superimpositions
Agent layer	Claude + Python	Coordinate tools, files, decisions and reports

MODELLER performs the structural calculation; Claude coordinates and documents the pipeline.

Planned outputs and visual analysis

09



3D comparison

Template, predicted model and experimental structure shown individually and as colored superimpositions

Results table

Target	Method	TM	GDT	IDDT	RMSD
T1124	MODELLER	-	-	-	-
	AlphaFold3	-	-	-	-
T1123	MODELLER	-	-	-	-
	AlphaFold3	-	-	-	-
T1127	MODELLER	-	-	-	-
	AlphaFold3	-	-	-	-

Final interpretation: connect prediction quality to template identity, coverage and alignment quality.

One-week schedule

